RiboMetric Report

Analysis of **SRR11005875.Aligned.toTranscriptome.sorted.bam** with **gencode.v49.annotation_RiboMetric.tsv**.
Completed 14:00:19 10/06/2026.

fail

This sample has material QC concerns; inspect the flagged metrics and plots before using it as a reference example. Top-line values: periodicity 79.1%, CDS enrichment 92.0%, recommended read proportion 80.6%.

Periodicity

79.1%

pass

CDS enrichment

92.0%

pass

Recommended reads

80.6%

pass

Duplicate rate

60.9%

fail

Multimapper rate

94.3%

fail

Metric Summary

Mapping

Metric	Summary	Score	Status
Duplicate rate	Estimated fraction of weighted reads explained by duplicate collapsed sequences.	60.9%	fail
Multimapper rate	Backwards-compatible weighted RPF-level fraction with evidence of multiple genomic loci.	94.3%	fail

RPF multimapper rate	Weighted fraction of ribosome protected fragments with evidence of multiple genomic loci.	94.3%	fail
Unique RPF rate	Weighted fraction of ribosome protected fragments considered uniquely mapped.	5.7%	fail
Alignment multimapper rate	Fraction of reported alignment rows whose fragment has evidence of another genomic alignment.	95.4%	fail
5' soft clips	Fraction of weighted reads with 5 prime soft clipping, often indicating trimming or adapter issues.	0.0%	pass

Footprint Lengths

Metric	Summary	Score	Status
Read length distribution iqr metric	Scores how tightly footprint lengths concentrate around the central distribution.	66.7%	warn
Read length distribution coefficient of variation metric	Scores footprint length consistency using the coefficient of variation.	93.0%	pass
Read length distribution maxprop metric	Fraction of reads in the most abundant footprint length.	38.6%	fail

Read length distribution bimodality metric	Scores whether the footprint length distribution is dominated by one mode rather than multiple modes.	81.1%	pass
Disome proportion	Fraction of reads in the configured di-ribosome footprint length window.	0.0%	pass
Recommended reads	Fraction of reads in lengths recommended for downstream frame-sensitive analysis.	80.6%	pass

Periodicity

Metric	Summary	Score	Status
Periodicity information	Information-content score for separation between reading frames.	75.7%	pass
Periodicity information weighted score	Read-depth weighted information-content score for reading-frame separation.	68.9%	warn
Uniformity entropy	Entropy-based score for how evenly reads cover coding regions.	99.7%	pass
Periodicity	Fraction of coding reads in the dominant reading frame after offset assignment.	79.1%	pass

Annotation / CDS

Metric	Summary	Score	Status
Stop codon readthrough ratio	Ratio of downstream to upstream signal around stop codons.	6.6%	pass
Start codon enrichment ratio	Read enrichment near start codons relative to downstream coding background.	2.56	pass
Cds coverage metric	Score summarising how much annotated CDS sequence is covered by reads.	43.9%	fail
Cds coverage	Observed proportion of annotated CDS positions covered by reads.	43.9%	fail
Cds coverage metric not inframe 1read 1000tx	Quality-control metric included in the configured RiboMetric summary.	11.9%	fail
Cds coverage not inframe 1read 1000tx	Quality-control metric included in the configured RiboMetric summary.	11.9%	fail
Cds coverage metric not inframe 100read 100tx	Quality-control metric included in the configured RiboMetric summary.	0.7%	fail
Cds coverage not inframe 100read 100tx	Quality-control metric included in the configured RiboMetric summary.	0.7%	fail
Cds coverage metric inframe 1read 1000tx	Quality-control metric included in the configured RiboMetric summary.	20.6%	fail

Cds coverage inframe 1read 1000tx	Quality-control metric included in the configured RiboMetric summary.	20.6%	fail
Cds coverage metric inframe 100read 100tx	Quality-control metric included in the configured RiboMetric summary.	1.8%	fail
Cds coverage inframe 100read 100tx	Quality-control metric included in the configured RiboMetric summary.	1.8%	fail
CDS enrichment	Fraction of annotated reads assigned to CDS regions.	92.0%	pass
Prop reads leader	Fraction of annotated reads assigned to leader or 5 prime UTR regions.	3.2%	fail
Prop reads trailer	Fraction of annotated reads assigned to trailer or 3 prime UTR regions.	4.6%	fail

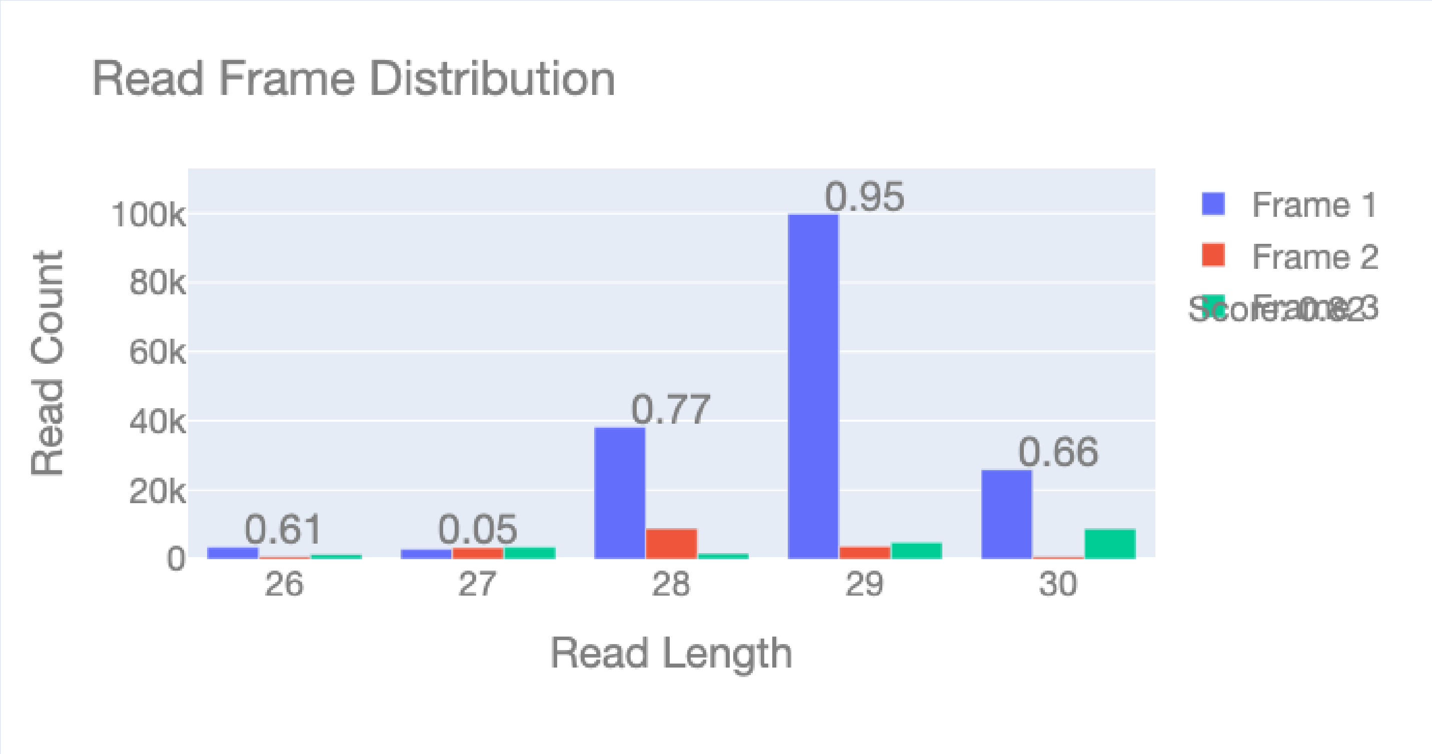
Codon / Sequence

Metric	Summary	Score	Status
Terminal nucleotide bias distribution 5 prime metric	Scores nucleotide balance at the 5 prime end of reads.	84.6%	pass
Terminal nucleotide bias distribution 3 prime metric	Scores nucleotide balance at the 3 prime end of reads.	88.1%	pass

Terminal bias kl 5prime	KL-divergence based score for 5 prime terminal nucleotide bias.	84.6%	pass
Terminal bias kl 3prime	KL-divergence based score for 3 prime terminal nucleotide bias.	88.1%	pass
Terminal bias kl 5prime score	Quality-control metric included in the configured RiboMetric summary.	84.6%	pass
Terminal bias kl 3prime score	Quality-control metric included in the configured RiboMetric summary.	88.1%	pass
Terminal bias kl 5prime raw	Quality-control metric included in the configured RiboMetric summary.	18.2%	pass
Terminal bias kl 3prime raw	Quality-control metric included in the configured RiboMetric summary.	13.5%	pass
Terminal bias maxabs 5prime	Largest absolute 5 prime terminal nucleotide deviation from expectation.	86.3%	pass
Terminal bias maxabs 3prime	Largest absolute 3 prime terminal nucleotide deviation from expectation.	91.7%	pass

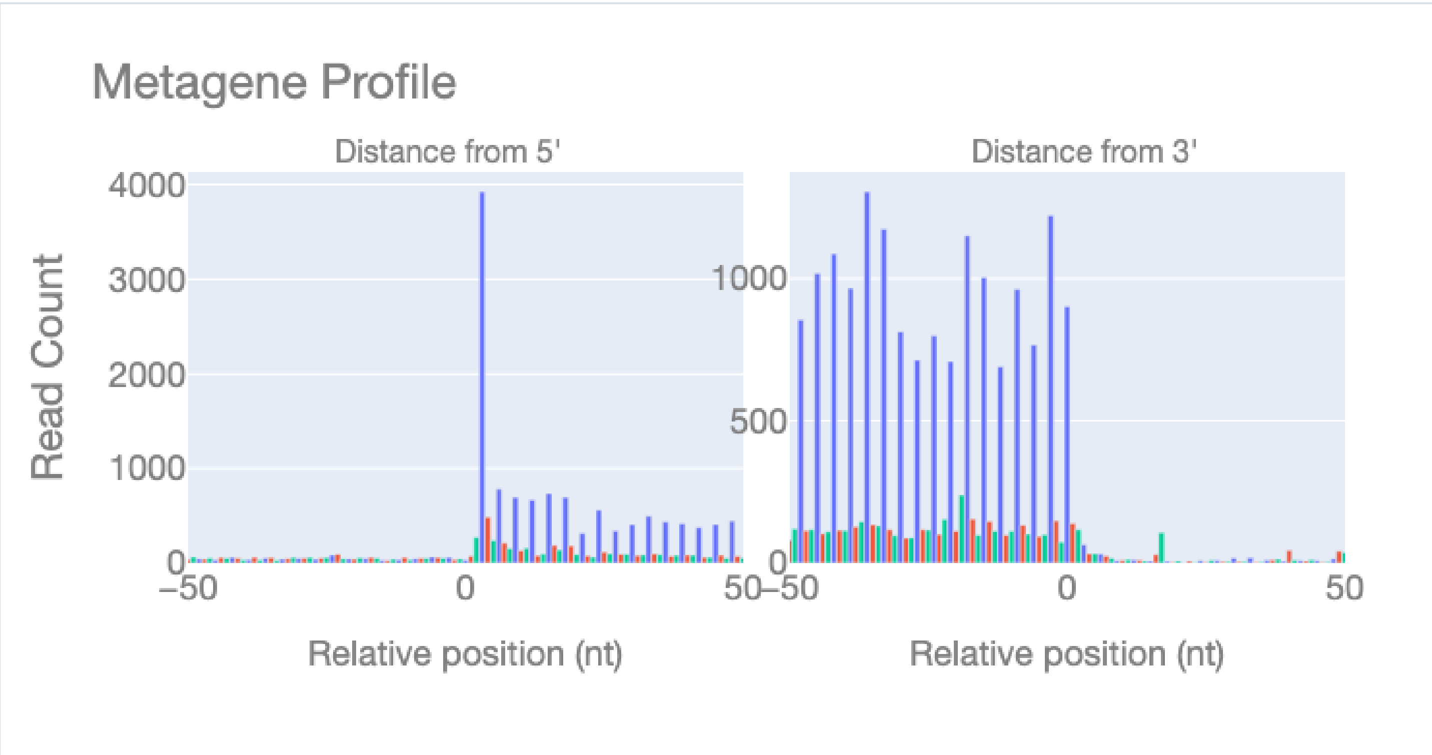
Read Frame Distribution

Frame distribution per read length



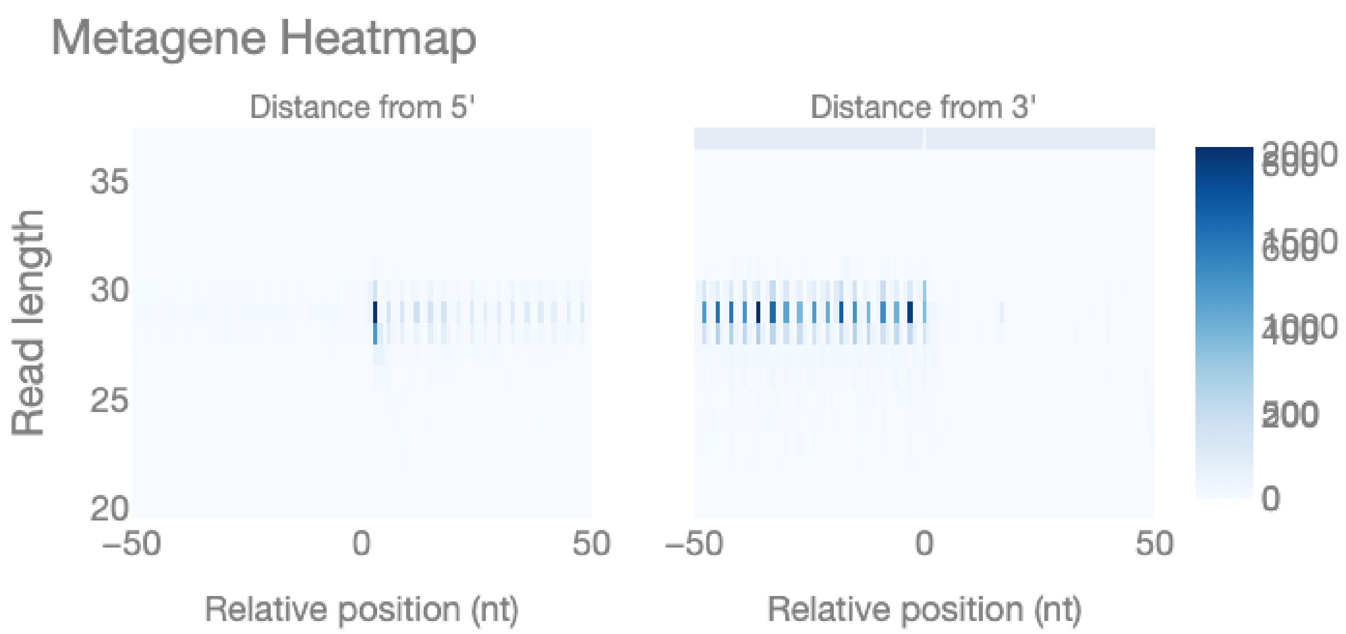
Metagene Profile

Metagene profile showing read counts by distance from the selected target. The plotted read population follows the multimap filter. MAPQ=255 unique reads; profile reads n=264,839/6,715,027; window reads: start n=18,135, stop n=21,158.



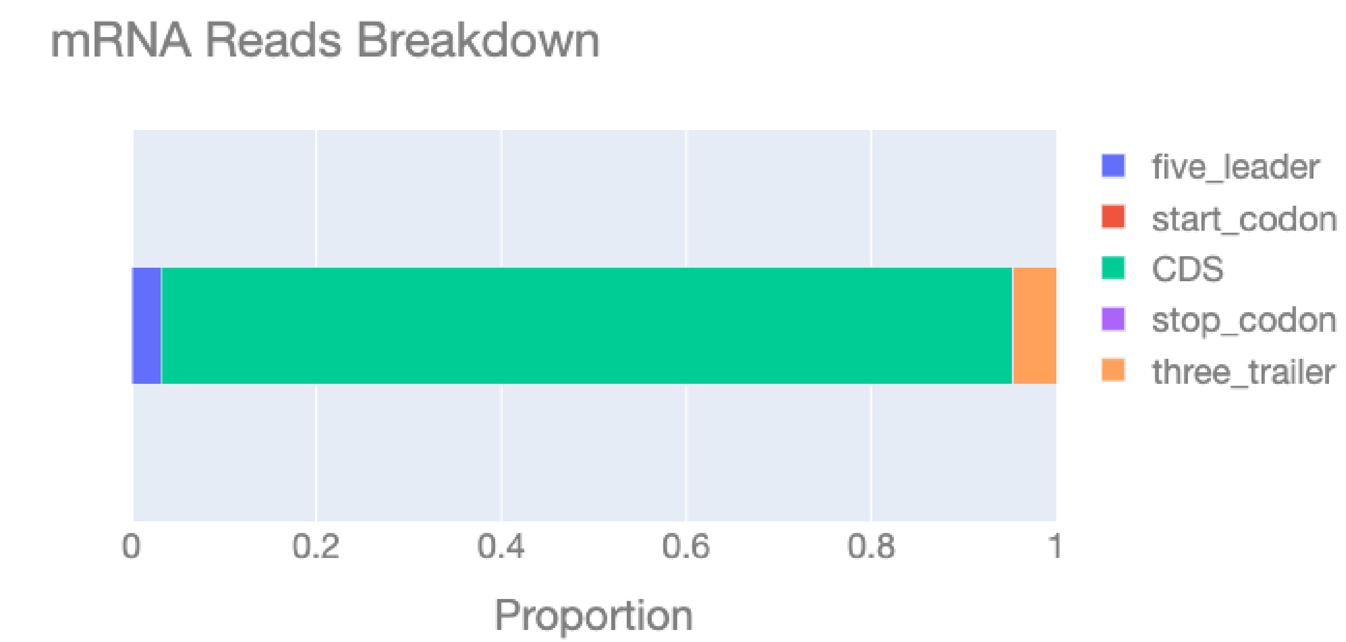
Metagene Heatmap

Metagene heatmap showing A-site distance from the selected target by read length. The plotted read population follows the multimap filter. MAPQ=255 unique reads; profile reads n=264,839/6,715,027; window reads: start n=18,135, stop n=21,158.



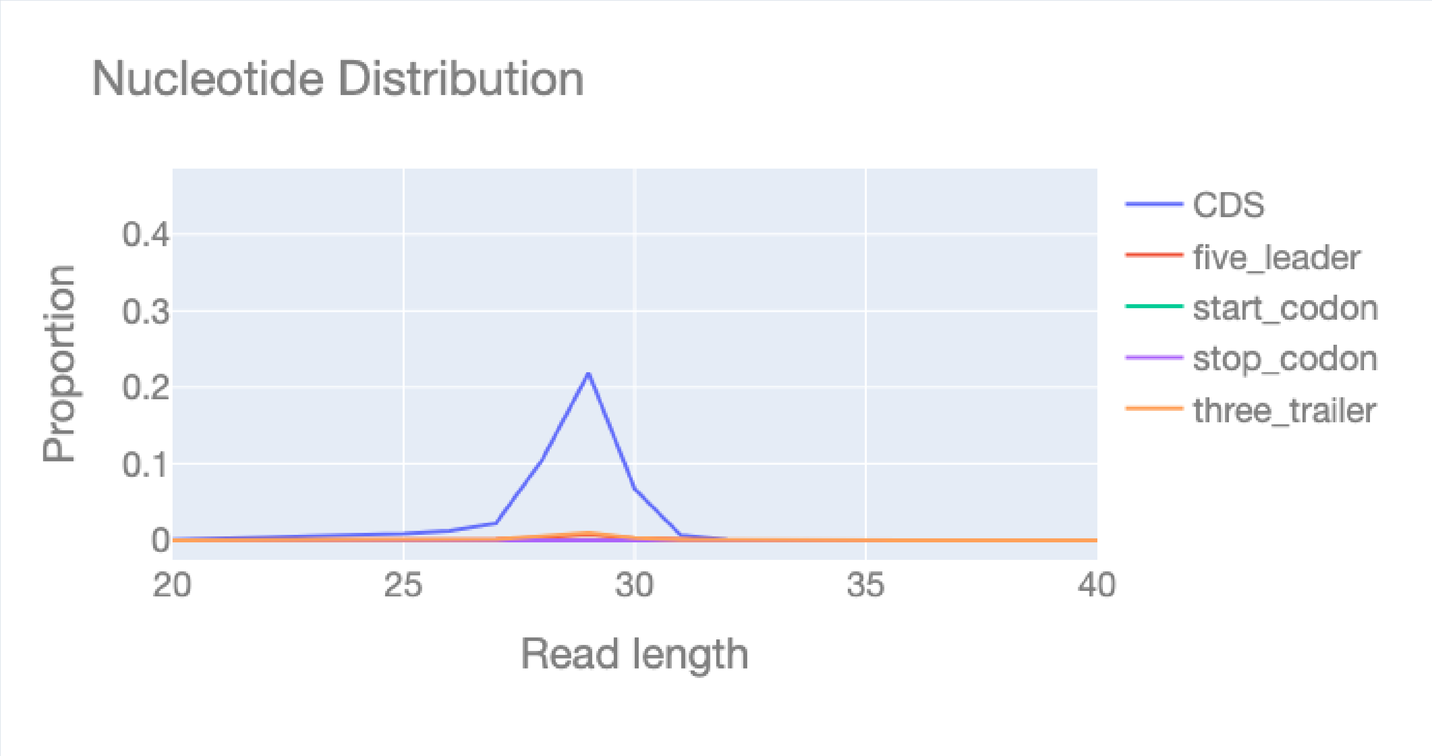
mRNA Reads Breakdown

Shows the proportion of the different transcript regions represented in the reads



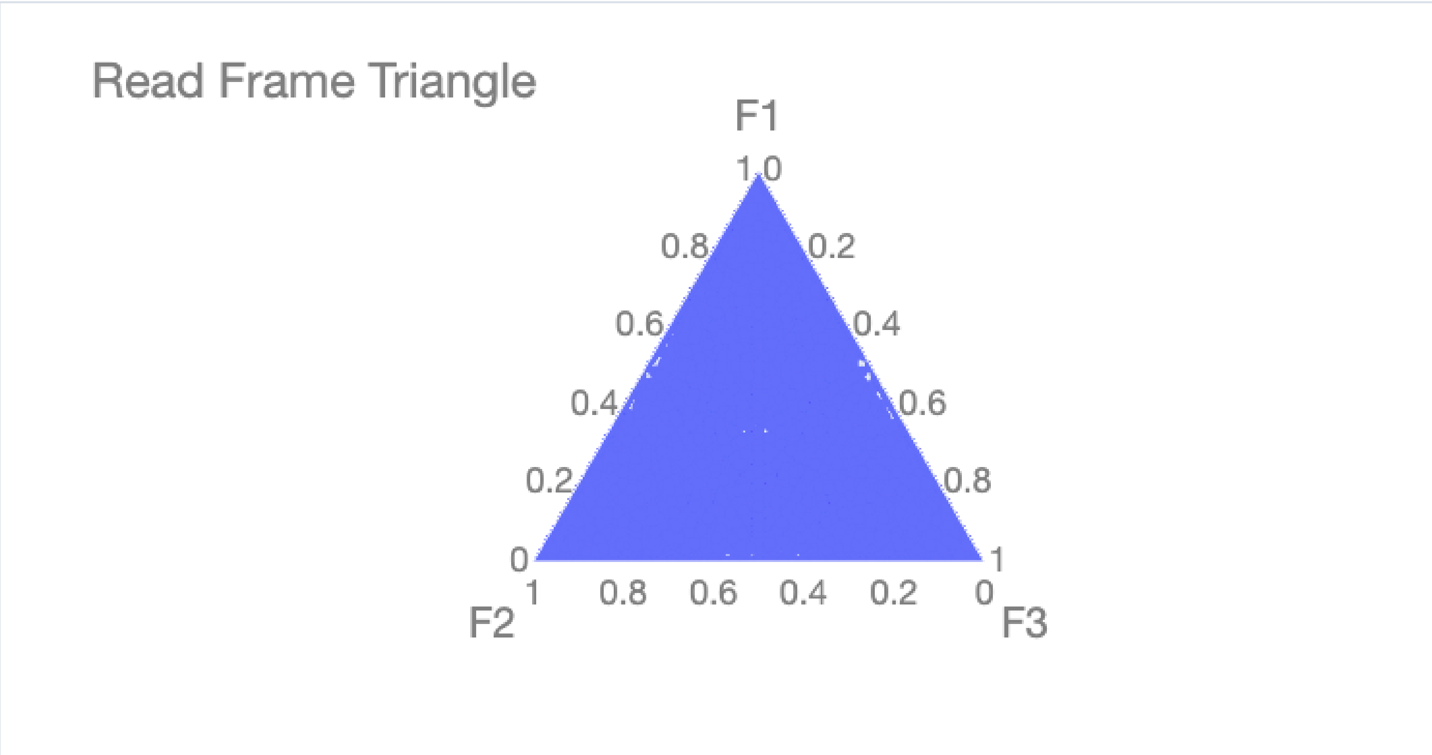
mRNA Reads Breakdown over Read Length

Shows the proportion of the different transcript regions represented in the reads over the different read lengths.



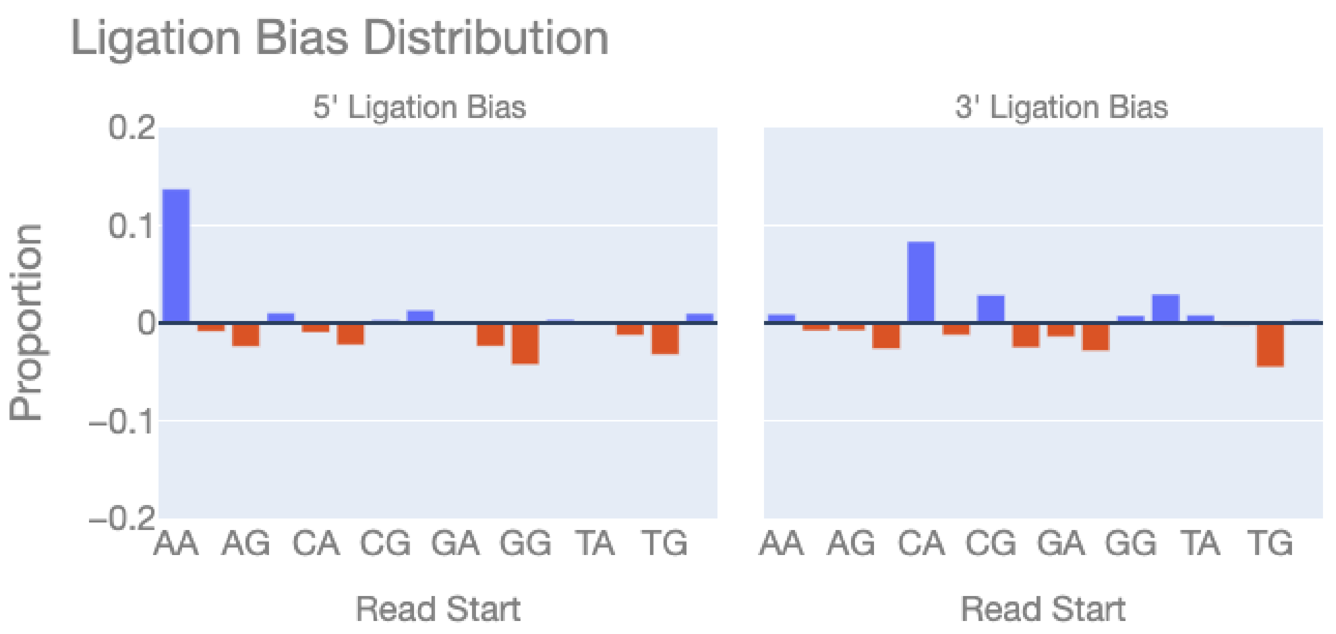
Read Frame Triangle

Triangle plot showing the distribution of read frames for the full dataset



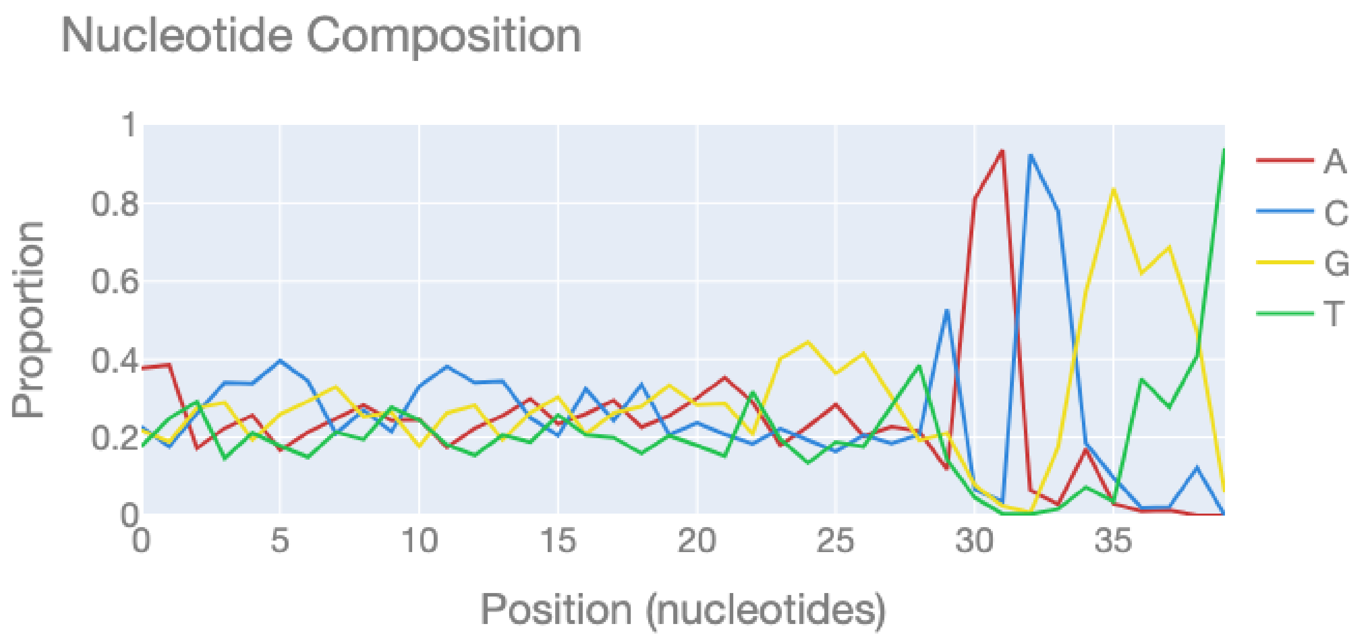
Ligation Bias Distribution

Distribution of end bases for the full dataset



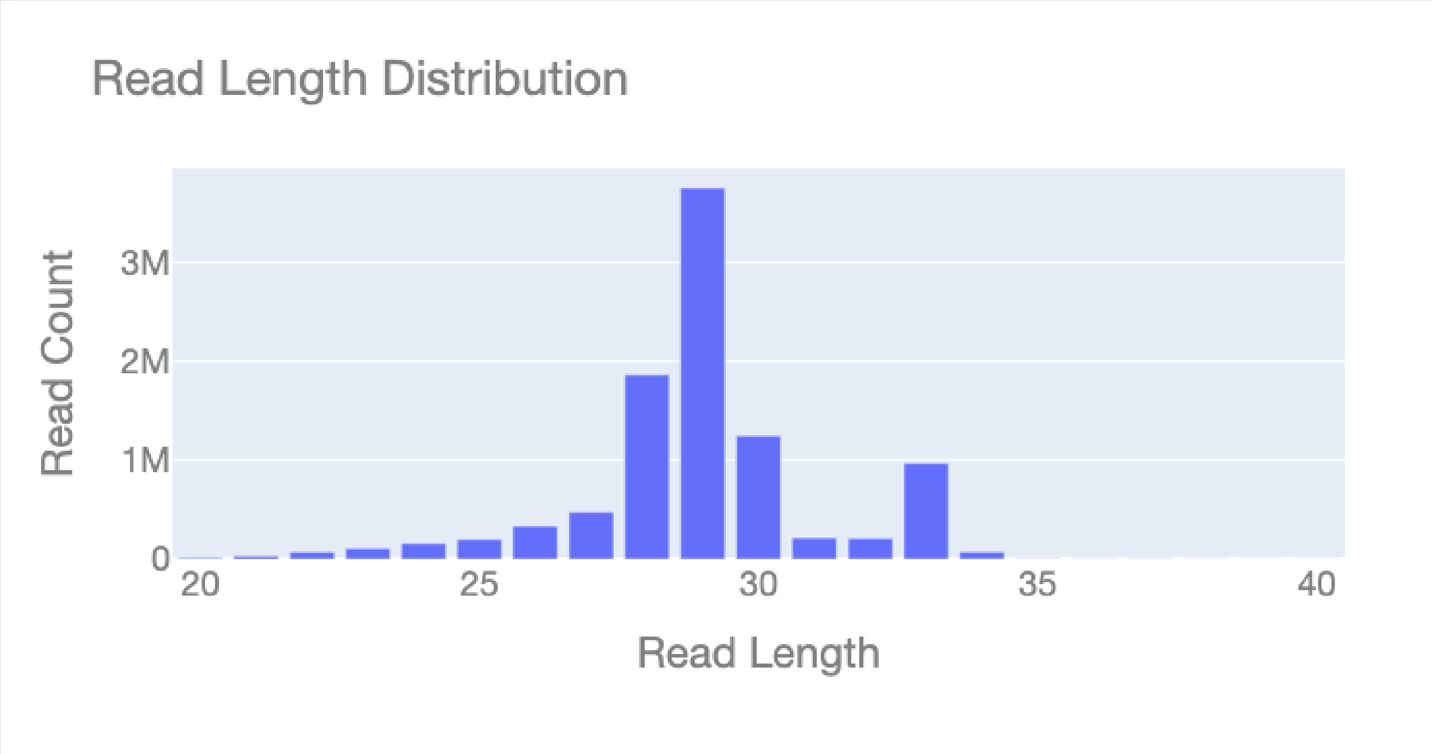
Nucleotide Composition

Nucleotide composition of the reads



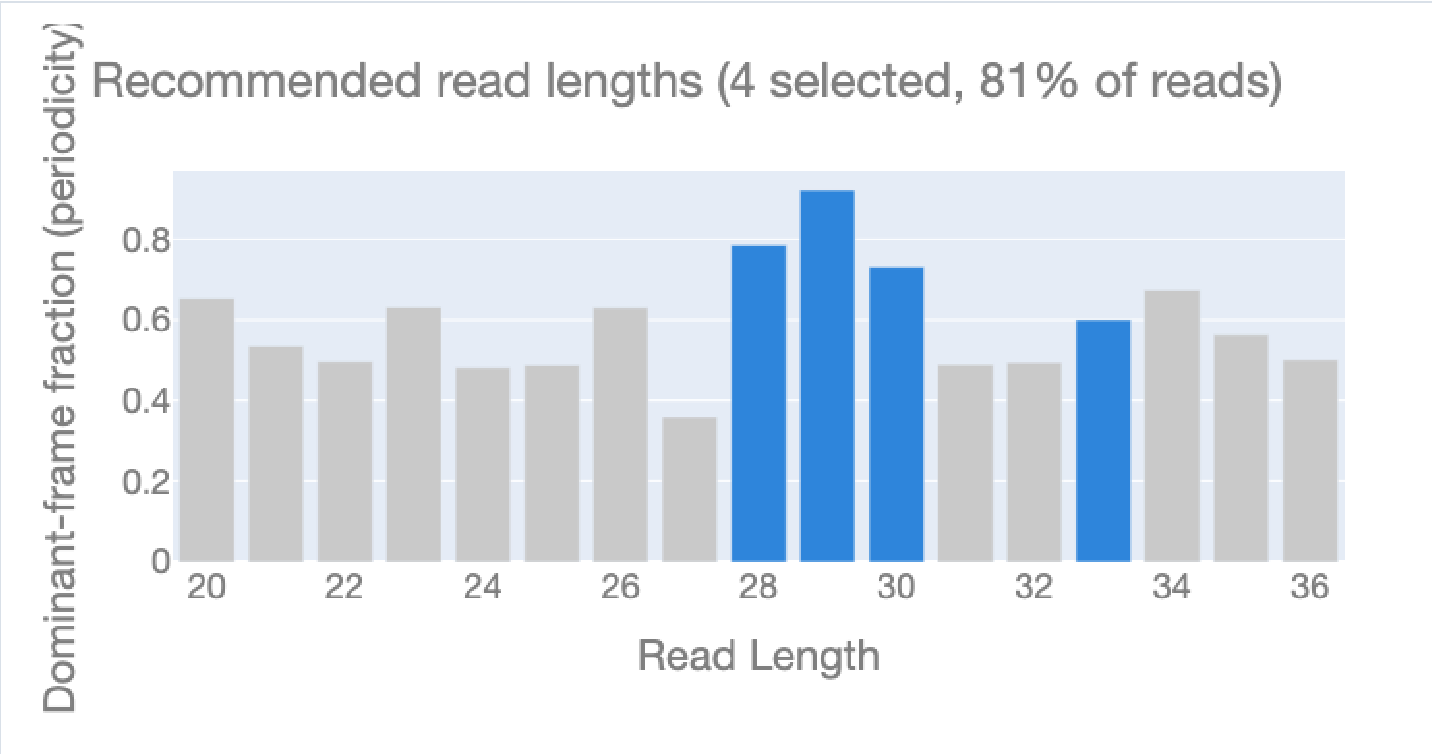
Read Length Distribution

Distribution of read lengths for the full dataset



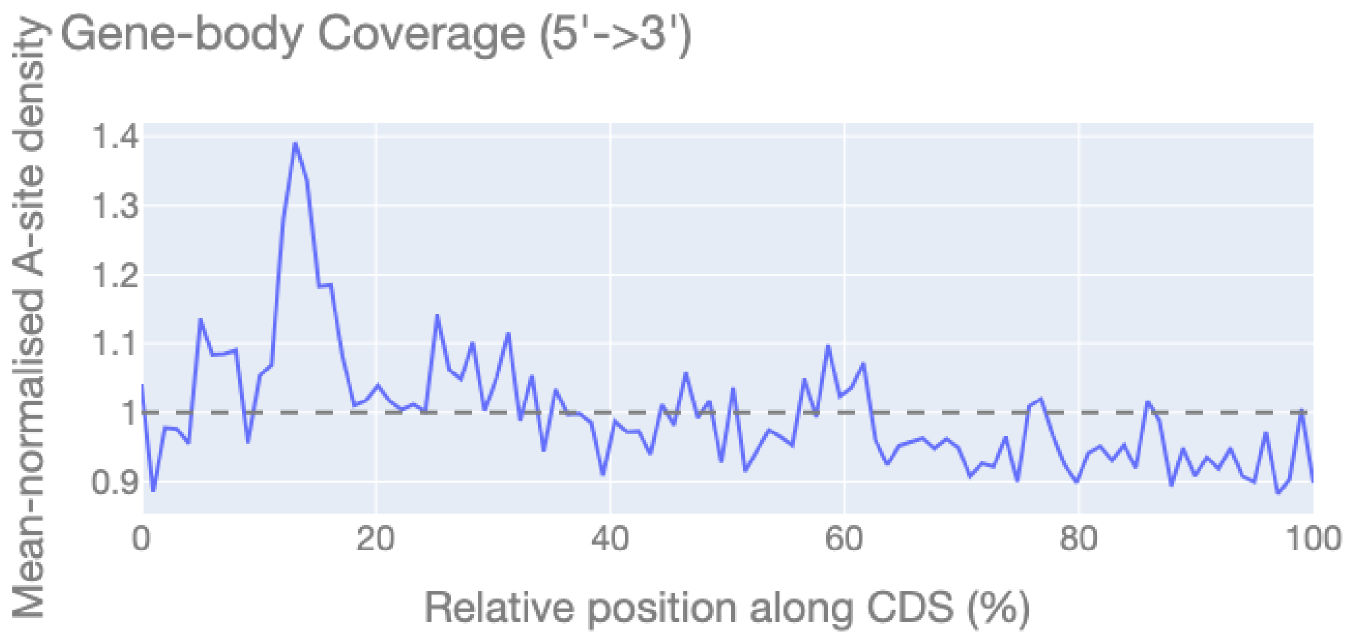
Recommended Read Lengths

Per-read-length 3-nt periodicity. Blue bars are recommended for downstream P-site assignment / ORF calling (periodic enough and carrying enough reads); grey bars are not.



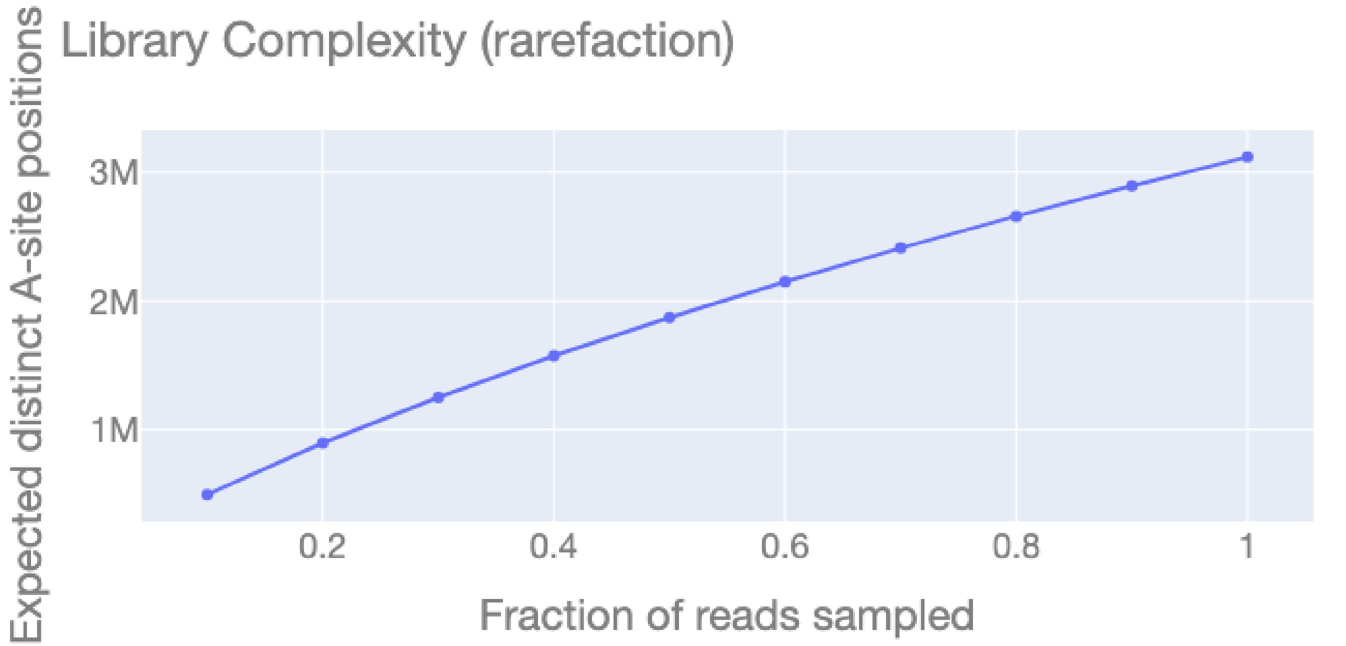
Gene-body Coverage

A-site density across relative CDS position. A 5' bump is the translation ramp; a 3' decline indicates drop-off. A flat line at 1.0 is perfectly uniform elongation.



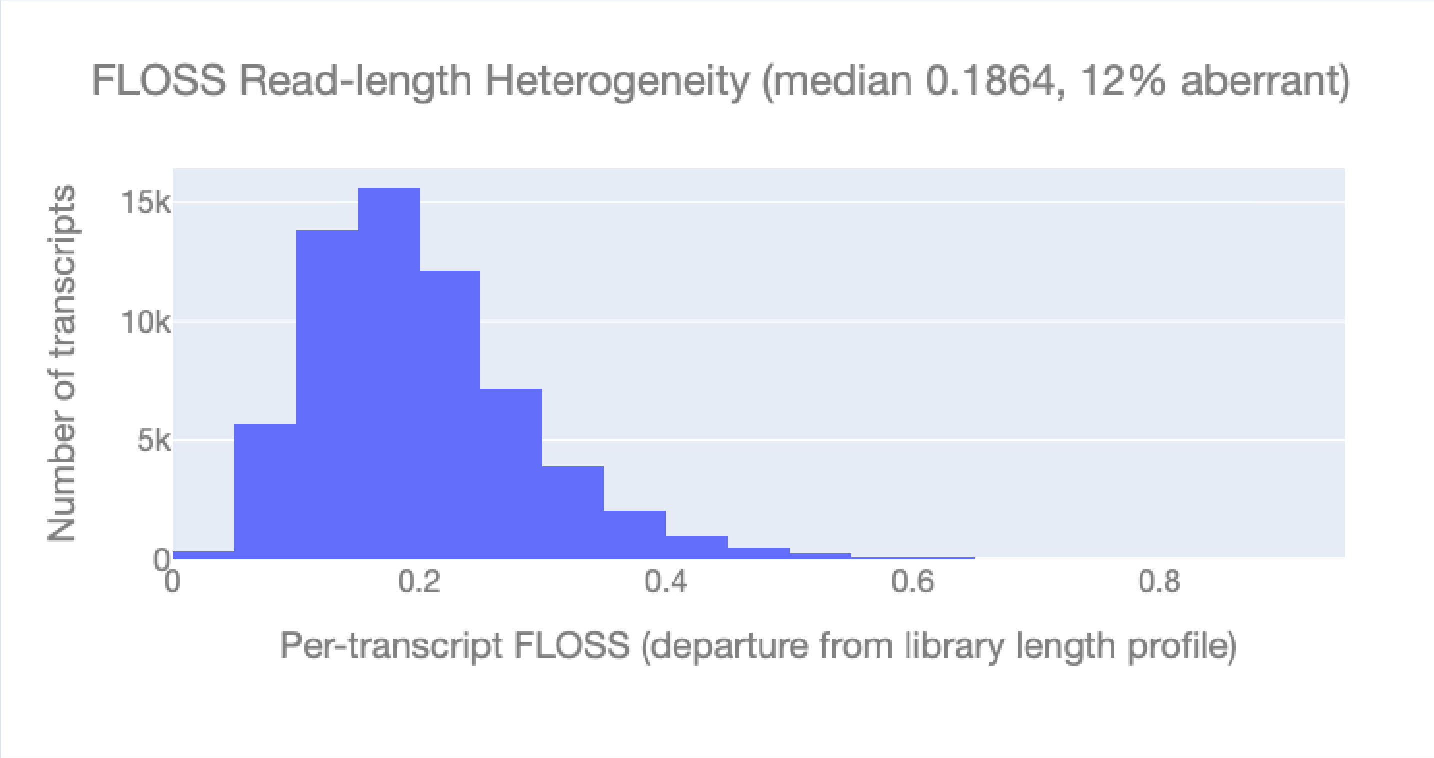
Library Complexity

Expected distinct A-site positions recovered vs sequencing depth. A curve that has flattened by full depth is saturated (extra sequencing adds little); a still-rising curve is under-sequenced.



FLOSS Heterogeneity

Distribution of per-transcript FLOSS scores (footprint-length departure from the library aggregate). A long right tail means many transcripts have anomalous length profiles — a heterogeneous or contaminated library.


FLOSS Read-length Heterogeneity (median 0.1864, 12% aberrant)
Number of transcripts
15k
10k
5k
0
0
0.2
0.4
0.6
0.8
Per-transcript FLOSS (departure from library length profile)